OOP Project Final Report

Flash Quiz Application

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1 Introduction

The Flash Quiz application is a Java-based interactive tool designed to help users enhance their learning experience through structured revision and self-assessment. The application provides an intuitive interface for creating, managing and reviewing flashcards, as well as a quiz mode to test knowledge retention.

The primary goal of this application is to facilitate active recall, a scientifically proven learning technique that strengthens memory through repeated exposure to concepts. The Flash Quiz application is particularly beneficial for students, professionals, and lifelong learners who are looking for an efficient way to retain information.

2 Class Diagram:

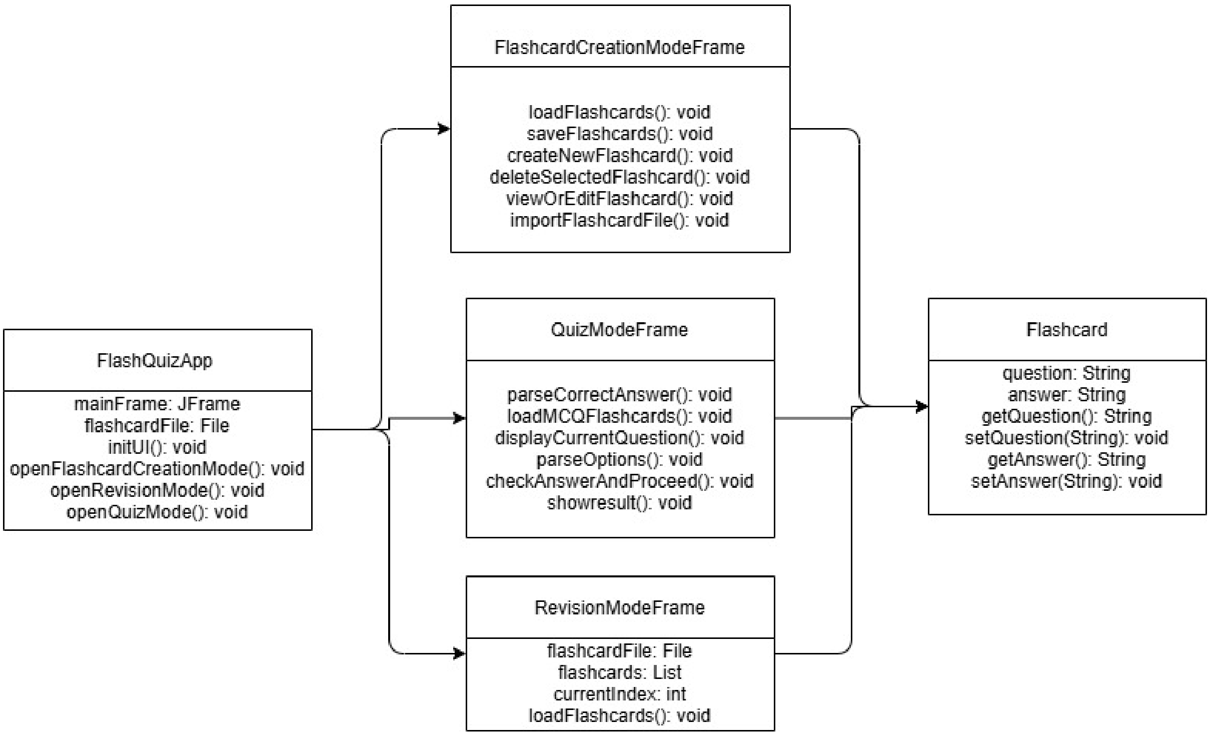

Figure 1: Class Diagram

3 Home Page

The Home Page serves as the central hub of the Flash Quiz Application, providing users with easy navigation options.

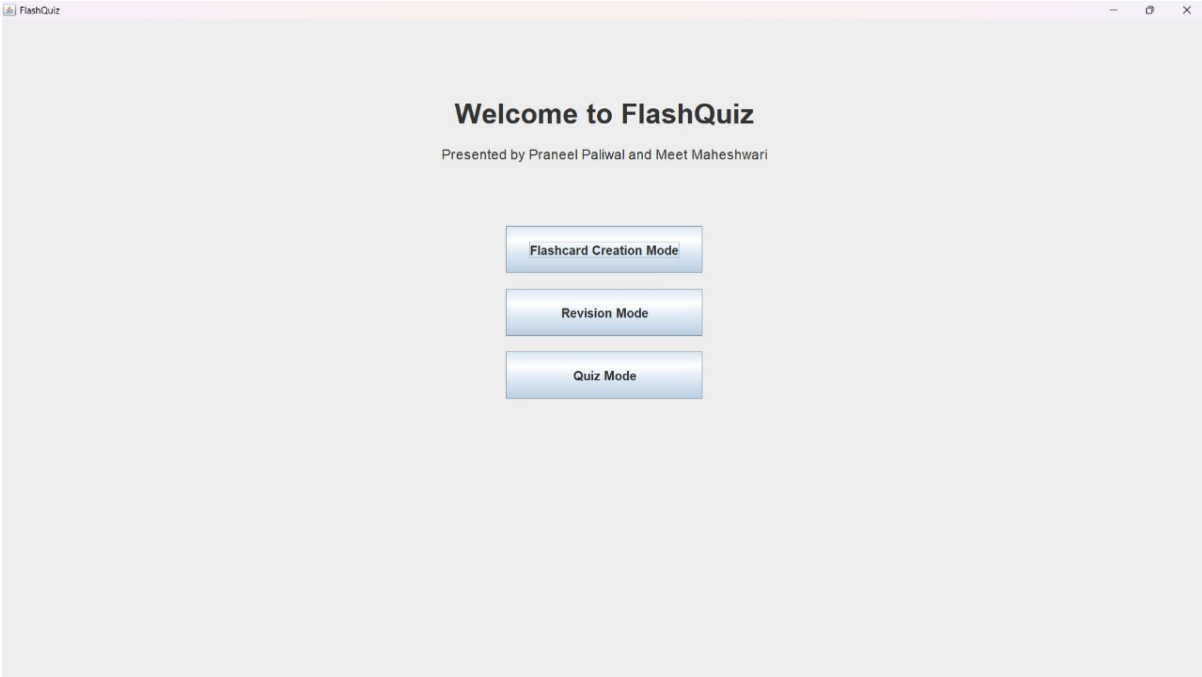

Figure 2: Welcome to FlashQuiz

It features a clean and user-friendly interface with buttons or menu options that lead to the core functionalities of the application.

- **Flashcard Creation:** Allows users to add, edit, or delete flashcards.
- **Revision Mode:** Allows structured review of flashcards.
- **Quiz Mode:** Facilitates knowledge assessment through interactive quizzes.

4 Flashcard Creation

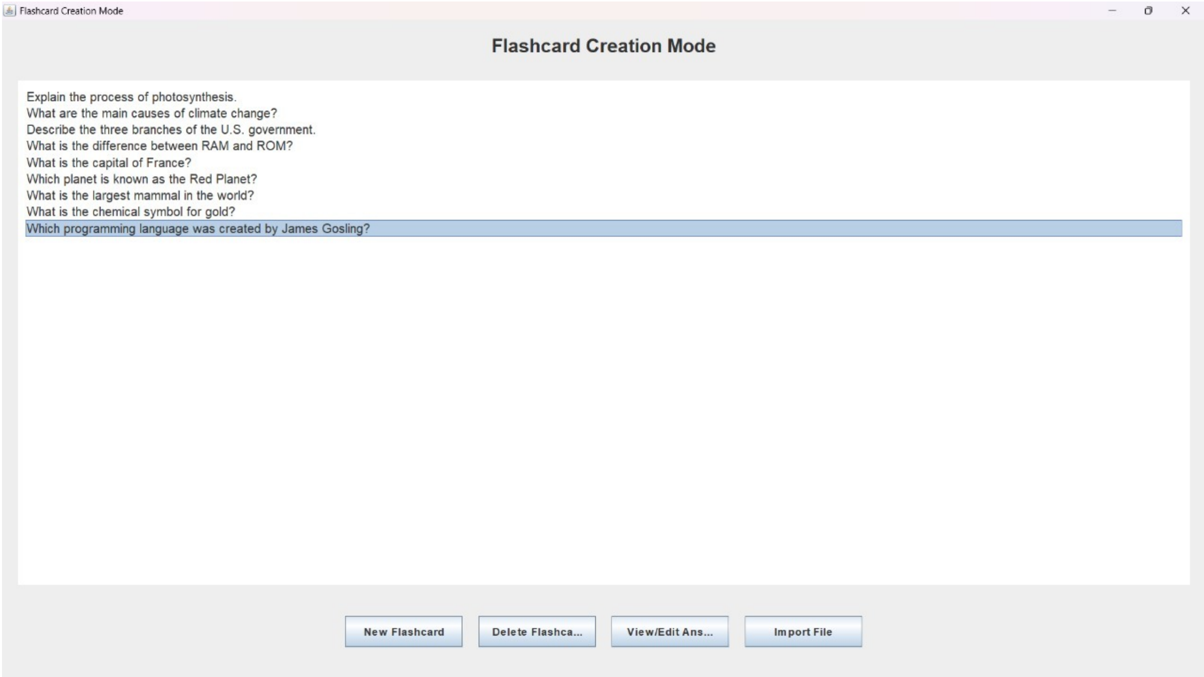

Figure 3: Flash Card Creation Mode

The Flashcard Creation module allows users to create and manage study materials.

1. By clicking the New Flashcard button, users see an input option as shown in Figure 4.

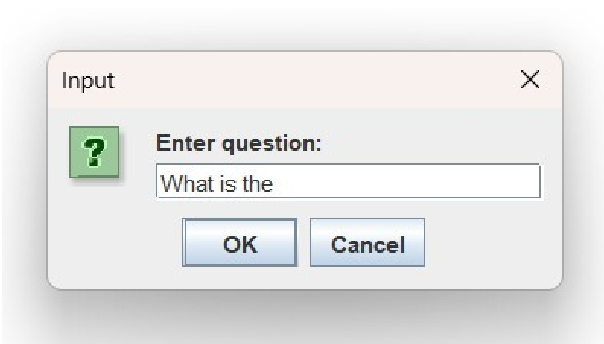

Figure 4: Question Input

2. After entering the question the user gets a prompt asking if it is a MCQ or a Subjective Question 5.

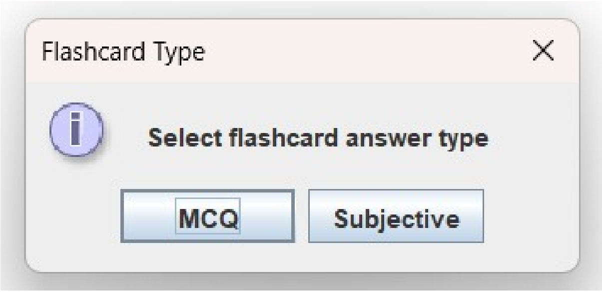

Figure 5: Answer Type Selection

3. Input for Subjective Answer 6.

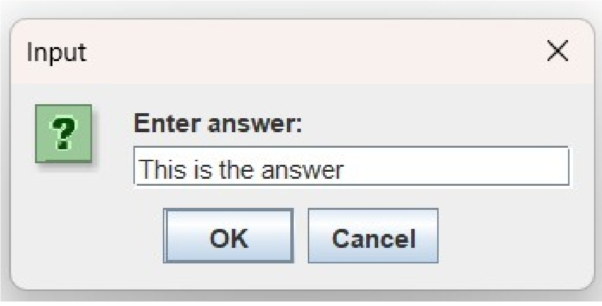

Figure 6: Answer Subjective

4. If we select MCQ the we are asked the Number of Choices: 7.

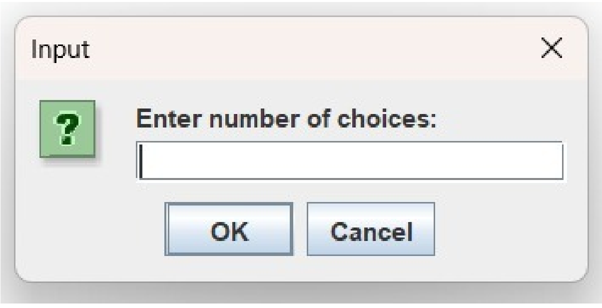

Figure 7: MCQ Number of Choices Input

5. Once the number of choices is entered then the user inputs the choices one by one 8.

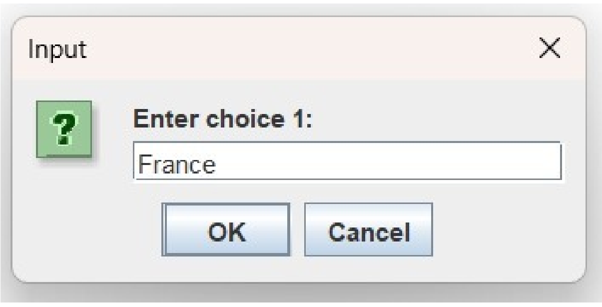

Figure 8: Choices Input

6. After all choices are inputted then the user enters the correct choice 9.

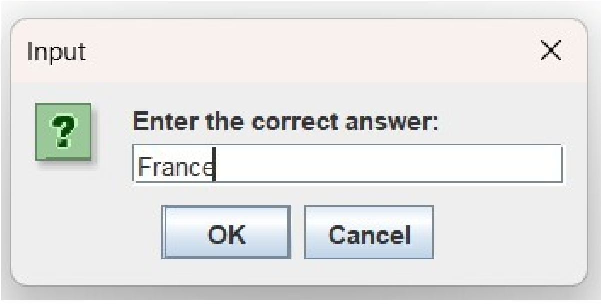

Figure 9: Input of the Correct Answer

7. There is another way to create flashcards by importing a specially formatted file 10.

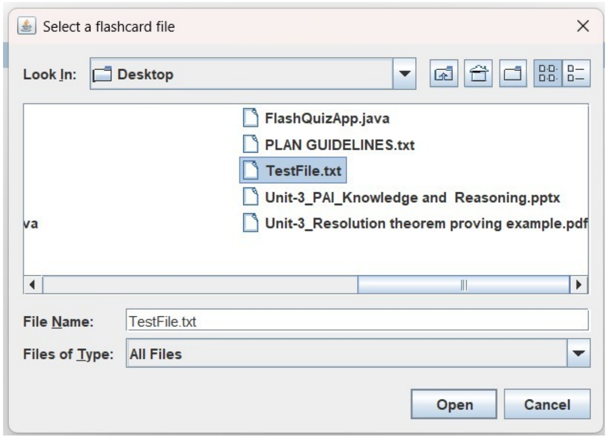

Figure 10: Input for Imported File

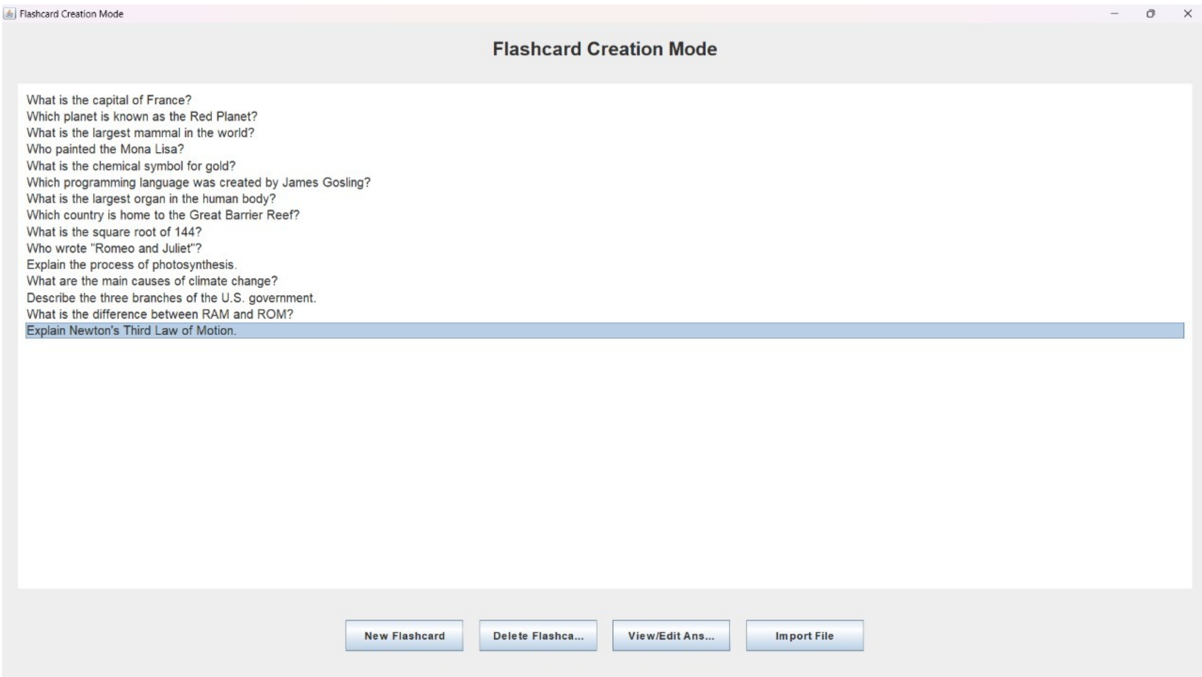

Figure 11: Data From Imported File

5 Revision Mode

Revision mode facilitates effective learning by enabling users to review their flashcards systematically.

In this mode the user can go through all the Flashcards such that first they see only the question and then on clicking show answer they will see the corresponding answer as show in 13

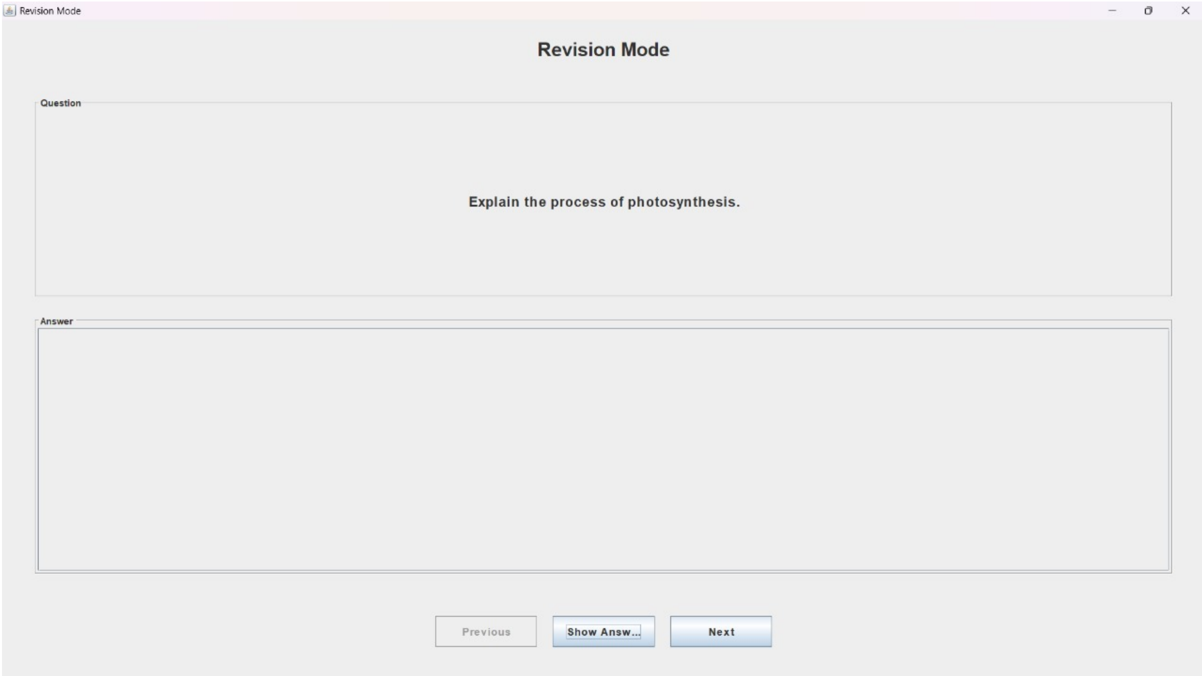

Figure 12: Revision Mode Interface

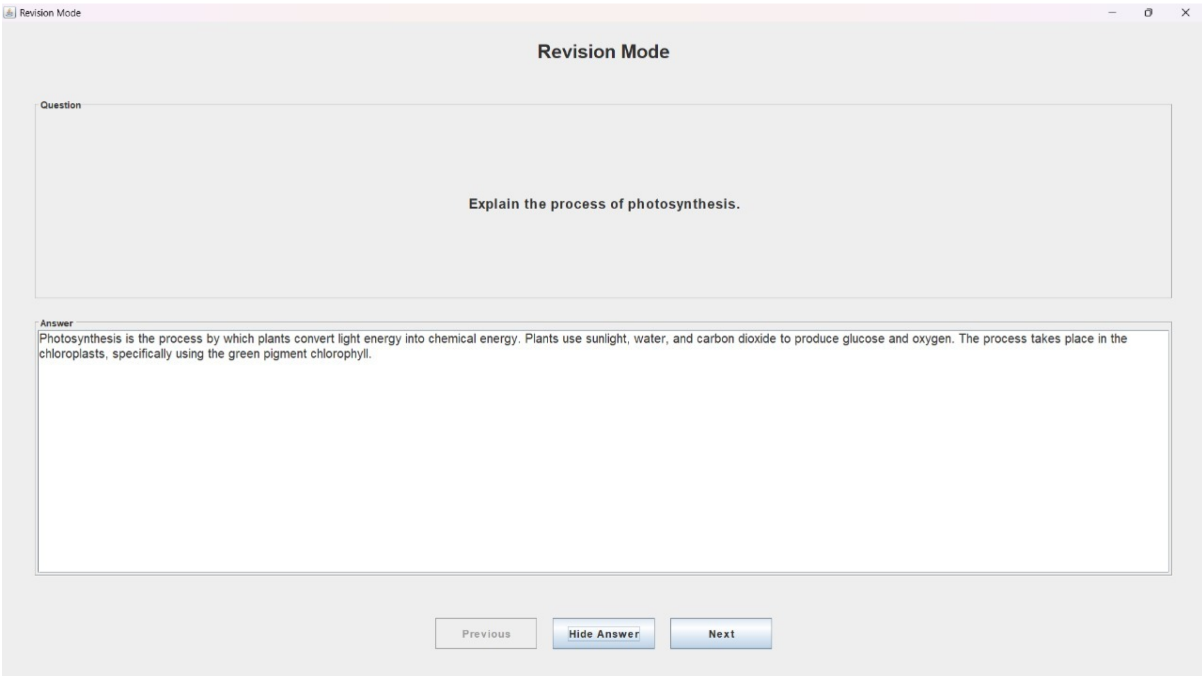

Figure 13: Revision Mode Interface

6 Quiz Mode

Quiz Mode serves as an assessment tool to test users' knowledge based on the flashcards they have created: The user sees the questions as in a manner as shown in 14:

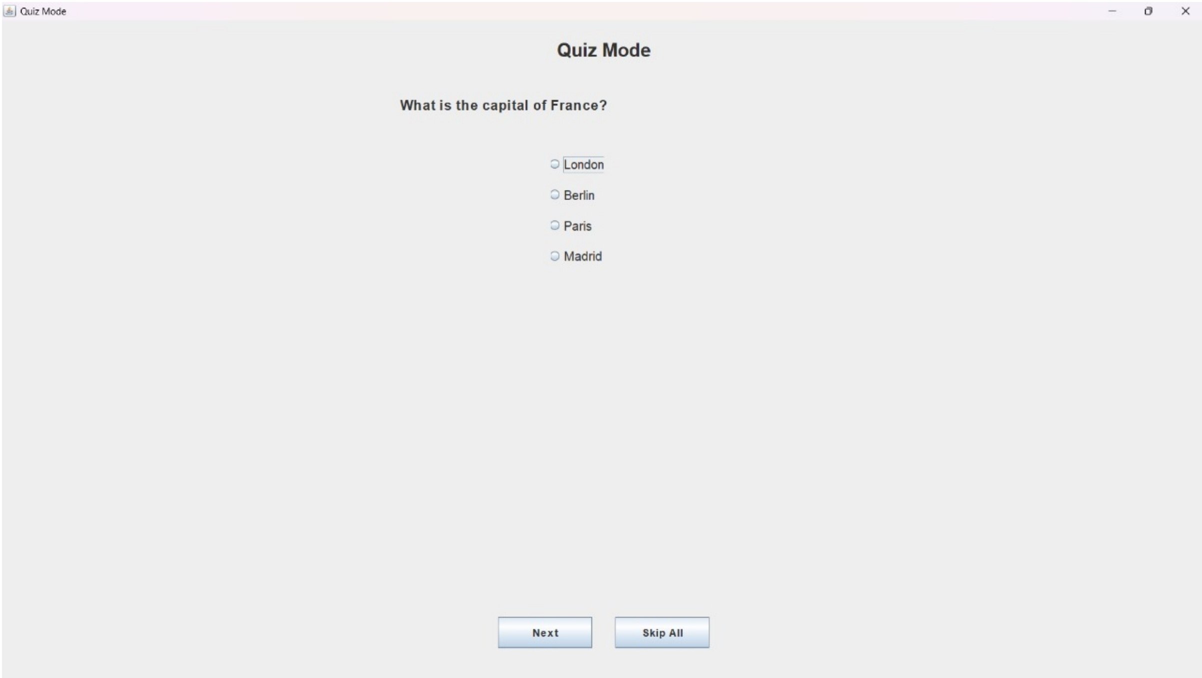

Figure 14: Quiz Mode

In the end, the user sees the result as shown in 15:

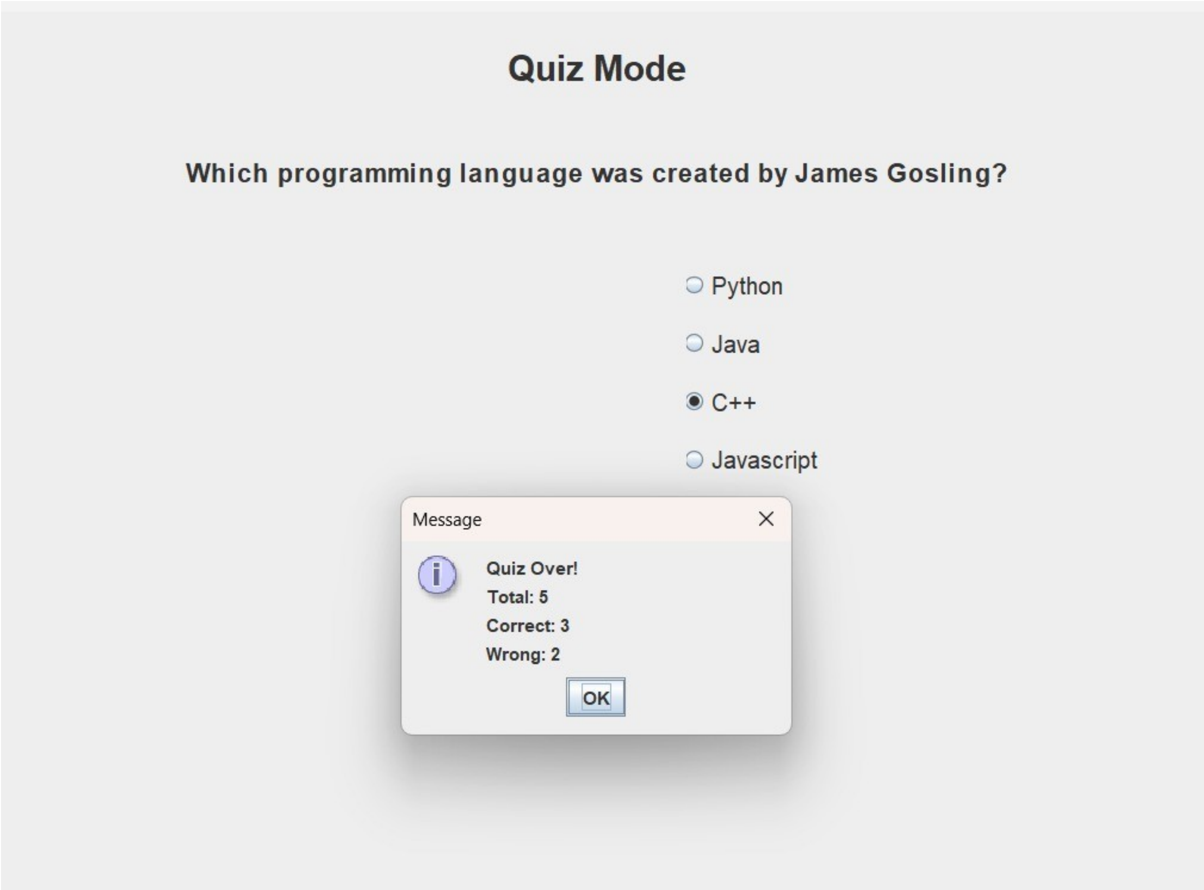

Figure 15: Quiz Mode

- **Question Types:** Supports multiple choice questions.
- **Instant Feedback:** Provides immediate results and explanations.
- **Immediate Feedback:** Displays quiz score so that the user can understand their weaknesses.

7 Backend

The backend functionality of the FlashQuizApp manages flashcards, handles file operations, and supports various modes of interaction using basic File Handling as shown in16 and 17.

7.1 File Handling

The application uses a file named `flashcards.txt` to persistently store flashcard data. The backend implements the following file operations:

1. **File Creation:** Upon initialization, the app checks for the existence of `flashcards.txt`. If it doesn't exist, a new file is created.
2. **Reading Flashcards:** The `loadFlashcards()` method uses a `BufferedReader` to read the file contents line by line and parses each line into `Flashcard` objects.
3. **Saving Flashcards:** The `saveFlashcards()` method utilizes a `PrintWriter` to write flashcard data to the file using a consistent format with "Q:" for questions and "A:" for answers.

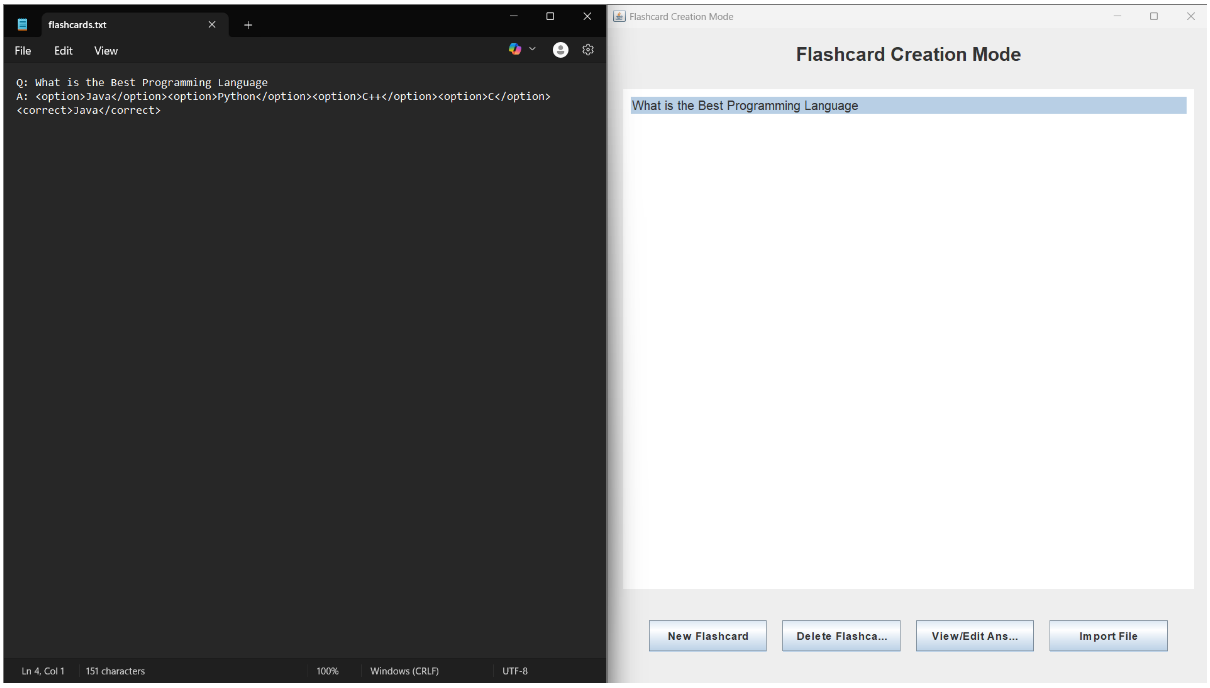

Figure 16: Backend File Handling Input

7.2 Flashcard Management

The backend includes a robust system for managing flashcards:

1. **Flashcard Class:** A custom `Flashcard` class encapsulates each flashcard with question and answer fields, along with getter and setter methods.
2. **Flashcard Storage:** All flashcards are stored in a `List` structure, enabling efficient manipulation and iteration.
3. **CRUD Operations:**
   - **Creation:** Users can add new flashcards via input.
   - **Reading:** Flashcards are displayed and browsable via the UI.
   - **Updating:** Flashcards can be edited after selection.
   - **Deletion:** Unwanted flashcards can be removed from the list.

7.3 Importing Flashcards

The backend supports importing flashcards from external text files:

1. **File Selection:** A `JFileChooser` allows the user to select a `.txt` file.
2. **Parsing:** The file is parsed line by line, extracting "Q:" and "A:" prefixes.
3. **Validation:** The format is validated, and exceptions (e.g., invalid files or IO errors) are handled gracefully.

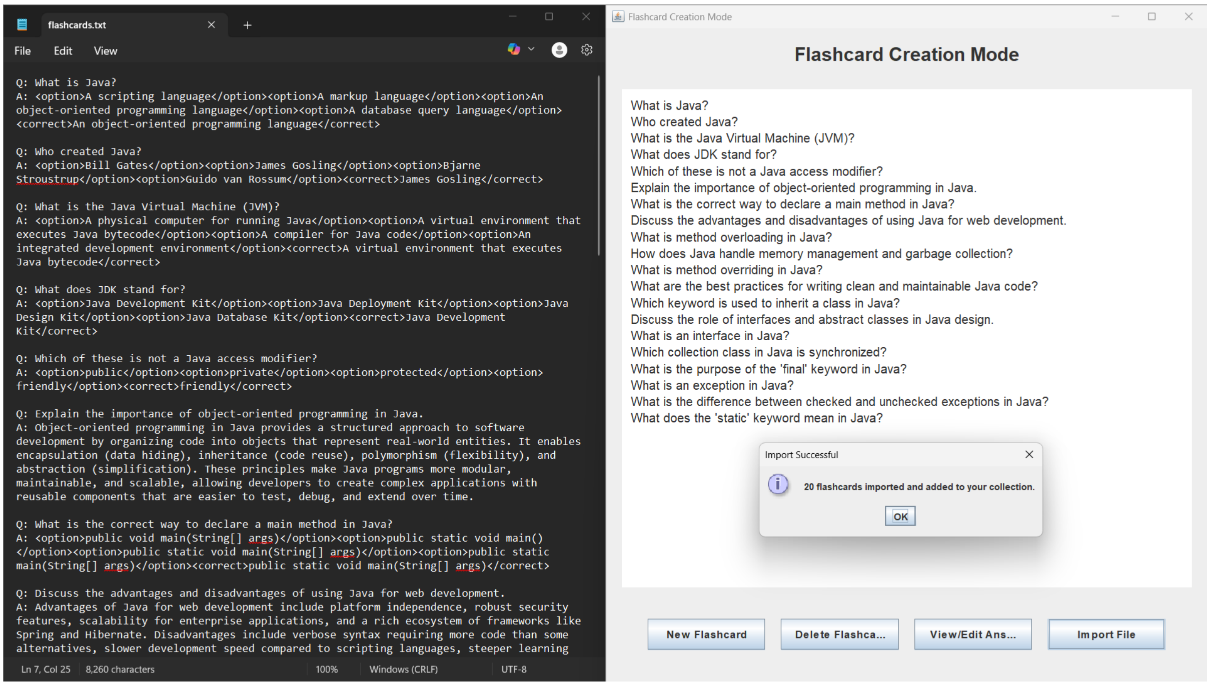

Figure 17: Backend File Handling Input

7.4 Revision Mode

Revision mode enables users to systematically review their flashcards:

1. **Flashcard Navigation:** Tracks the current flashcard index and provides methods to move to the next or previous flashcard.
2. **Answer Toggle:** Allows users to reveal or hide answers as needed.
3. **MCQ Handling:** Parses multiple-choice options from the answer string and formats them for display.

7.5 Quiz Mode

The quiz mode serves as an assessment tool:

1. **MCQ Filtering:** Filters the flashcard list to include only multiple-choice questions.
2. **Answer Checking:** Compares the user's selected answer with the correct answer.
3. **Score Tracking:** Tracks correct and incorrect answers, calculating the final score.

7.6 UI Interaction

The backend interacts closely with the UI:

1. **Event Handling:** Responds to user interactions like button clicks or list selections.
2. **Data Population:** Populates UI components with flashcard data.
3. **State Management:** Manages the state of UI elements (e.g., enabling/disabling buttons).

8 Conclusion

The Flash Quiz Application is a robust and innovative tool designed to streamline the process of learning and knowledge retention. By offering features such as flashcard creation, revision mode, and quiz mode, it caters to various learning styles and enhances user engagement. The application's ability to handle both subjective and multiple-choice questions ensures flexibility in content delivery. Future enhancements could significantly improve the usability and functionality of the application. These include:

- **Improving the User Interface (UI): A more intuitive and visually appealing interface will make navigation seamless and enhance the overall user experience.**
- **Adding Timed and Untimed Quizzes: Introducing quiz formats with time constraints or without them will cater to different learning preferences, allowing users to practice under exam-like conditions or at their own pace.**
- **Simplifying Input Methods for Multiple-Choice Questions (MCQs): Streamlining the process of creating MCQs will enable users to focus on content creation rather than technical complexities.**

Additionally, incorporating features such as gamification elements, progress tracking, and personalized feedback can further motivate users and improve learning outcomes. The ability to import flashcards from external files already demonstrates the adaptability of the application, which can be expanded further for collaboration with educational institutions or corporate training programs.